No of Pages : 4 **Course Code :23MX15**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

MCA Semester : 1

23MX15 WEB TECHNOLOGIES

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Course Outcome : Table

Qn.1	CO1	Qn.	CO2	Qn.3	CO3.	Qn.4	CO4

COs	At the end of learning this course, students will be able to
CO1	paraphrase and apply the fundamental internetworking concepts and technology needed for deploying web applications.
CO2	author web pages with well-structured markup languages and layout/styling patterns to create visually appealing static web pages.
CO3	create dynamic and interactive user interfaces that can respond to user input with built-in routines to provide real-time feedback.
CO4	identify , design , and apply appropriate tools in implementing solutions for event-driven web applications for social needs.

		Question	BTL
1.	a)	Match the following service/protocol to the related layer/service/ protocol. (3) i) Reliable protocol - a. Datalink Layer ii) Stateless protocol - b. Connection oriented iii) Hop-to-Hop delivery - c. TCP iv) frames to bits - d. Transport layer v) Remote Login - e. Physical layer vi) Socket address - f. HTTP	[L2]
	b) i)	What is the primary purpose of a web crawler in search engines? How do metadata like keywords, description help crawlers? (3)	[L2]
	ii)	Justify the statement “HTTP is a stateless protocol”. Also explain any four idempotent HTTP methods. (7)	[L3]
	c) i)	Explain the complete rendering process in a browser, starting from receiving the HTML file to displaying the visual page on the screen. (12)	[L3]
		(OR)	
	ii)	How do web caches, cookies, and search engines work together to optimize user experience and website performance during a repeated visit to a previously searched webpage and in a new search?	[L2]

2. a) Write the difference between block-level and inline elements in HTML. Give an example where using an inline element is more appropriate than a block-level element. (3) [L3]

b) i) How is CSS attribute position:sticky different from position:fixed? Also define a style for positioning the <h1> element 50px from the top, relative to the HTML page. (3) [L4]

ii) Create a HTML table for a billing system that includes item details and calculates the total bill. The table will include item names, quantity, price per unit, total price for each item, and finally, the total amount for all items. (7) [L5]

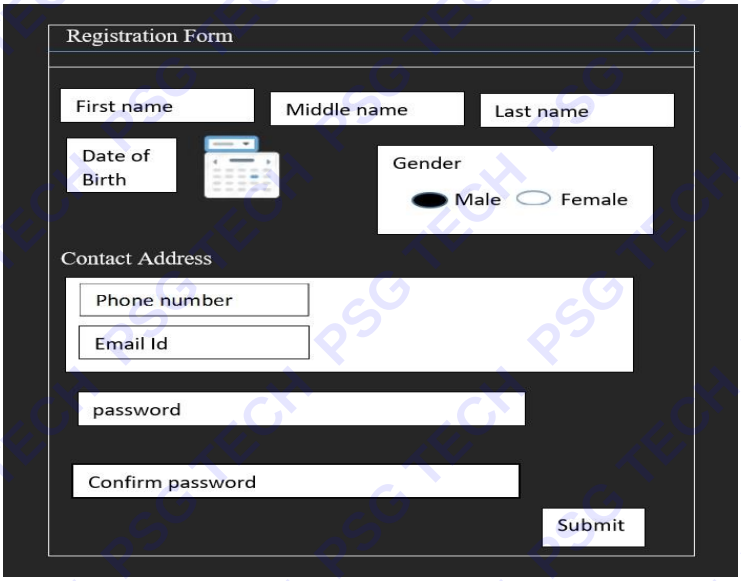
Bill Details:

Item Name	Quantity (Kg)	Price per kg (Rs.)	Total price (Rs.)
Apple	2	160.00	320.00
Banana	3	80.00	240.00
Total	5		560.00

Also write the CSS selectors for applying the following styles:

- Zebra Striping: Even rows alternate with a light gray background, which improves readability.
- Hover Effect: When you hover over a row, it changes to a slightly darker gray, indicating interactivity.
- Total Row Styling: The Total Bill row is styled with a distinct orange background to separate it from other rows.

c) Develop a HTML code for generating the following registration form with the following fields. Include proper validation and CSS styles. On click of submit button, if the validation succeeds, your application should display "Successful registration " message in the next page. (12) [L5]



Using pattern attribute of HTML validate the input fields for following conditions:

- Name: Must contain only letters and spaces.
- Date of Birth: Must be a valid date. Should allow people born on or before 2004 only.
- Email Address: Must be a valid email.
- Phone Number: Must match the 10 digit pattern for a valid phone

number (e.g., 1234567890).

- Restrict the user to set strong password with one lowercase and uppercase letters, a number, a special symbol with a minimum of totally 8 characters.

3. a) State the purpose of stringify and parse methods of JSON and Predict the Output of the following JS code snippet: (3) [L4]

```
const students = [
  { name: 'John',age: 21},
  { name:'Akbar',age: 22},
  { name:'Kumar',age: 23} ];
  let data= JSON.stringify(students);
  console.log(data,"\n");
  constobj = JSON.parse(data);
  console.log(obj);
```

b) i) Fill up the missing regular expression statement in the following JavaScript code so as to generate the output as specified in the comments lines. (4) [L5]

```
function validateEmail(email) {
  // Fill in the regex pattern below
 const regex = /________/;
  return regex.test(email);
}
// Test cases
console.log(validateEmail("user@example.com")); // true
console.log(validateEmail("user.nam@domain.co"));//true
console.log(validateEmail("user@.com")); // false
console.log(validateEmail("user@domain")); // false
console.log(validateEmail("user123@domain.org"));// true
console.log(validateEmail("user@domain@another.com")); // false
```

ii) List out the significance of Document Object Model in web development. Create a webpage that contains a button and a paragraph. When the button is clicked, the text of the paragraph should change to "Employee details updated!". Write the HTML and JavaScript needed to achieve the same. (6) [L3]

c) Explain the dynamic DOM and its role in web development. Discuss how DOM APIs such as querySelectorAll, createElement, and appendChild are used to dynamically manipulate the DOM structure. Provide examples of how these methods can be used together to create and manage dynamic web content. (12) [L4]

4. a) Analyze the following JavaScript code. If the variable xx in line number 11 is set to 58 what will be the output? If the same variable xx is set to 98 what will be the output? (3) [L4]

```
1. function taskOne(data) {
2.     try { if (data >75)
3.       console.log("Distinction");
4.       else if ( (data >60) && (data<75))
console.log("Excellent");
```

```
      else throw new Error("Need Improvement ");
    } catch (error) {
      console.log(`${error.message}`);  }
     finally {  console.log("Attempt Ends");  }
console.log("End of Calculation");
 }
let xx= __ ;
 try {
taskOne(xx);
   console.log("Calculated");
 } catch (error) {
   console.log(` ${error.message}`);
 } finally {
   console.log("Course Ends");
 }
```

b i) Predict the order of execution of statements in the following JavaScript code. Also give reasons. (5) [L5]

```
function delay(ms) {
    return new Promise((resolve,msg)=> {
setTimeout(()=>{  console.log("Acquire  Job  or Give Job");
      resolve();},ms )   }   )}

async function myFunction() {
      await delay(3000);
      console.log("end");   }
console.log("start");
myFunction();
setTimeout(  ()=>  {console.log("Focus  on Interview Preparation")},2000);
setTimeout(  ()=>  {console.log("Enrich  Soft Skills")},4000);
console.log("Aim of my life")
```

ii) Explain the difference between synchronous and asynchronous file operations in JS. [L3]

Write the proper JavaScript code to write a JSON object to a file using asynchronous file methods and read & display it back to the console. (5)

c) i) Describe the NODEJS server architecture and its working. Also explain how the NodeJS event loop architecture supports JavaScript in handling multiple tasks asynchronously. (12) [L4]

(OR)

ii) Develop a simple application of a user registration system. The system should allow users to register with user name and email id. Also add a feature to view all the registered users. [L5]

Explain middleware used and the routes needed for registering and viewing users.

CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL

FD/RL /END/